

## 1 $\pi$ attenuator

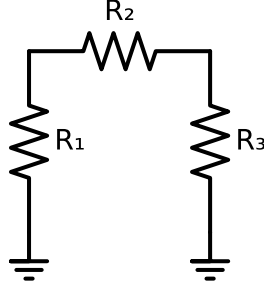


Figure 1:  $\pi$ -attenuator

The three constraints for the attenuator,

1.  $R_1 = R_3$
2.  $S_{11} = 0$ , or  $Z_{in} = R_0$
3.  $S_{21} = A$

Starting with the second constraint, when looking into the left side of the network, and assuming the right side is terminated in  $R_0 = 50\Omega$ ,  $Z_{in}$  is,

$$Z_{in} = (R_3 || R_0 + R_2) || R_1$$

$$Z_{in} = \frac{\left(\frac{R_3 R_0}{R_3 + R_0} + R_2\right) R_1}{\left(\frac{R_3 R_0}{R_3 + R_0} + R_2\right) + R_1}$$

$$Z_{in} = \frac{\left(R_3 R_0 + R_2(R_3 + R_0)\right) R_1}{\left(R_3 R_0 + R_2(R_3 + R_0)\right) + R_1(R_3 + R_0)}$$

Using  $R_1 = R_3$ , and the constraint that  $Z_{in} = R_0$ ,

$$R_0 = \frac{\left(R_1 R_0 + R_2(R_1 + R_0)\right) R_1}{\left(R_1 R_0 + R_2(R_1 + R_0)\right) + R_1(R_1 + R_0)}$$

Solve for  $R_2$ ,

$$R_0 \left( \left( R_1 R_0 + R_2(R_1 + R_0) \right) + R_1(R_1 + R_0) \right) = \left( R_1 R_0 + R_2(R_1 + R_0) \right) R_1$$

$$R_0 R_2(R_1 + R_0) - R_1 R_2(R_1 + R_0) = R_1^2 R_0 - R_1 R_0^2 - R_0 R_1(R_1 + R_0)$$

$$R_2 \left( R_0(R_1 + R_0) - R_1(R_1 + R_0) \right) = R_1^2 R_0 - R_1 R_0^2 - R_0 R_1(R_1 + R_0)$$

$$R_2 = \frac{R_1^2 R_0 - R_1 R_0^2 - R_0 R_1 (R_1 + R_0)}{R_0 (R_1 + R_0) - R_1 (R_1 + R_0)}$$

$$R_2 = \frac{2R_1 R_0^2}{R_1^2 - R_0^2}$$

The third constraint requires the voltage transfer function to be equal to the desired voltage loss,  $A$ .

$$S_{21} = \left. \frac{V_2^-}{V_1^+} \right|_{V_2^+ = 0} = A$$

Since the input impedance is matched,  $V_1^- = 0$ , and because the s-parameter definition requires that  $V_2^+ = 0$ ,

$$A = \frac{V_2}{V_1}$$

If a voltage  $V_1$  is applied at port 1, the voltage at port 2 is,

$$V_2 = V_1 \frac{R_3 || R_0}{R_2 + R_3 || R_0}$$

$$\frac{V_2}{V_1} = \frac{R_3 || R_0}{R_2 + R_3 || R_0}$$

$$\frac{V_2}{V_1} = \frac{\frac{R_3 R_0}{R_3 + R_0}}{R_2 + \frac{R_3 R_0}{R_3 + R_0}}$$

$$\frac{V_2}{V_1} = \frac{R_3 R_0}{R_2 (R_3 + R_0) + R_3 R_0}$$

Again using  $R_1 = R_3$ ,

$$A = \frac{V_2}{V_1} = \frac{R_1 R_0}{R_2 (R_1 + R_0) + R_1 R_0}$$

Solve for  $R_2$ ,

$$AR_2(R_1 + R_0) + AR_1 R_0 = R_1 R_0$$

$$R_2 = \frac{R_1 R_0 - AR_1 R_0}{A(R_1 + R_0)}$$

$$R_2 = \frac{R_1 R_0 (1 - A)}{A(R_1 + R_0)}$$

We have two unknowns,  $R_1$  and  $R_2$ , with two equations,

$$R_2 = \frac{2R_1 R_0^2}{R_1^2 - R_0^2}$$

$$R_2 = \frac{R_1 R_0 (1 - A)}{A(R_1 + R_0)}$$

Equating,

$$\frac{R_1 R_0 (1 - A)}{A(R_1 + R_0)} = \frac{2R_1 R_0^2}{R_1^2 - R_0^2}$$

Solving for  $R_1$ ,

$$\begin{aligned} R_1 R_0 (1 - A)(R_1^2 - R_0^2) &= 2R_1 R_0^2 A(R_1 + R_0) \\ (1 - A)(R_1^2 - R_0^2) &= 2R_0 A(R_1 + R_0) \\ (1 - A)(R_1^2) - 2R_0 A(R_1) &= 2R_0^2 A + R_0^2(1 - A) \\ (1 - A)(R_1^2) - 2R_0 A(R_1) - R_0^2(A + 1) &= 0 \\ (1 - A)(R_1^2) - 2R_0 A(R_1) + R_0^2(1 - A) &= 0 \end{aligned}$$

Using the quadratic formula to solve for  $R_1$ ,

$$R_1 = \frac{2R_0 A \pm \sqrt{4A^2 R_0^2 - 4R_0^2(1 - A)(A + 1)}}{2(1 - A)}$$

$$R_1 = \frac{R_0 A \pm R_0 \sqrt{A^2 - (A^2 + 1)}}{(1 - A)}$$

$$R_1 = R_0 \frac{A \pm 1}{1 - A}$$

$A$  is less than 1 (since attenuator is passive), so the denominator is always positive. For  $R_1$  to be non-negative, we must take the  $A + 1$  solution,

$$R_1 = R_0 \frac{A + 1}{1 - A}$$

Using this in the equation for  $R_2$ ,

$$R_2 = \frac{R_1 R_0 (1 - A)}{A(R_1 + R_0)}$$

$$R_2 = \frac{R_0 \frac{A+1}{1-A} R_0 (1 - A)}{A(R_0 \frac{A+1}{1-A} + R_0)}$$

$$R_2 = \frac{R_0(A + 1)}{A(\frac{A+1}{1-A} + 1)}$$

$$R_2 = R_0 \frac{(A + 1)(1 - A)}{A(A + 1 + (1 - A))}$$

$$R_2 = R_0 \frac{1 - A^2}{2A}$$